








Advanced Time Series Analysis and Forecasting

Chapter 6: State space models and Bayesian filtering



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **APF**: Auxiliary Particle Filter
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BN**: Beveridge–Nelson (decomposition)
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BSTS**: Bayesian Structural Time Series
- **DFM**: Dynamic Factor Model
- **DK**: Durbin–Koopman (simulation smoother, 2002)
- **DSGE**: Dynamic Stochastic General Equilibrium (model)
- **EKF**: Extended Kalman Filter
- **EM**: Expectation–Maximisation (algorithm)
- **EODHD**: EOD Historical Data (market-data provider)
- **ETS**: Error, Trend, Seasonality (exponential smoothing state space models)
- **FERMIAC**: mechanical Monte Carlo device of Enrico Fermi (Los Alamos, 1947)
- **FFBS**: Forward Filtering, Backward Sampling
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices
- **HP**: Hodrick–Prescott (filter)

Acronyms used in this material (2/2)

- **IG**: inverse gamma (distribution)
- **IMA**: Integrated Moving Average (model)
- **KSC**: Kim–Shephard–Chib (1998), mixture sampler for stochastic volatility
- **LLM**: Large Language Model
- **LR**: Likelihood Ratio (test)
- **MA**: Moving Average
- **MCMC**: Markov Chain Monte Carlo
- **ML**: Machine Learning
- **MNZ**: Morley–Nelson–Zivot (2003)
- **OLS**: Ordinary Least Squares
- **PMMH**: Particle Marginal Metropolis–Hastings
- **QML**: Quasi-Maximum Likelihood
- **SMC**: Sequential Monte Carlo
- **SV**: Stochastic Volatility
- **TSA**: Time Series Analysis
- **TVP**: Time-Varying Parameter(s)
- **TVP-VAR**: Time-Varying-Parameter Vector AutoRegression
- **TVP-VAR-SV**: TVP-VAR with Stochastic Volatility
- **UC**: Unobserved Components (model)
- **UC-SV**: Unobserved Components model with Stochastic Volatility (Stock–Watson 2007)
- **UC-UR**: UC model with correlated trend and cycle shocks (Morley–Nelson–Zivot 2003)
- **UC0**: UC model with uncorrelated trend and cycle shocks (Clark 1987)
- **UKF**: Unscented Kalman Filter
- **US**: United States
- **VAR**: Vector AutoRegression
- **VAT**: Value Added Tax

Today's question and route

- **Question**: how do we learn about quantities we never observe (trend inflation, the output gap, volatility, time-varying coefficients) and how much should we trust what we learn?
  - one framework: a state equation for the hidden quantity, a measurement equation for the data, and a filter that updates beliefs as data arrive
- **Route** of the chapter
  - the general linear Gaussian model, exact diffuse initialisation, the exact likelihood and inference at the boundary
  - Bayesian state space: simulation smoothers and Gibbs sampling; stochastic volatility
  - nonlinear filters: extended and unscented Kalman filters, particle filters, particle MCMC
  - time-varying parameters, dynamic factors, trend–cycle decompositions and trend inflation with stochastic volatility
- We build on TSA, Chapter 10 (state space form, Kalman filter by hand, smoothing, local level and trend, the UC output gap, Markov switching basics); Seminar 6 comes before this lecture

Learning outcomes

- Derive the Kalman filter and smoother for the general linear Gaussian model, with exact diffuse initialisation, and write them in `numpy`
- Maximise the exact likelihood and judge inference on variances that may sit on the boundary (pile-up, bootstrap LR tests)
- Sample states with the Carter–Kohn, Durbin–Koopman and precision samplers inside a Gibbs sampler; estimate stochastic volatility by the KSC mixture
- Run extended, unscented and particle filters; use the particle likelihood in PMMH and choose the number of particles
- Estimate TVP regressions, UC models with correlated shocks and UC-SV trend inflation, and report how fragile the latent estimates are

Reading, data and tools

- Backbone: Durbin and Koopman (2012), Ch. 2–9 and 11–14; Harvey (1990); Kim and Nelson (1999); Hamilton (1994), Ch. 13; Chopin and Papaspiliopoulos (2020)
  - original papers: Kalman (1960), Koopman (1997), Carter and Kohn (1994), Durbin and Koopman (2002), Kim, Shephard and Chib (1998), Gordon, Salmond and Smith (1993), Pitt and Shephard (1999), Andrieu, Doucet and Holenstein (2010), Primiceri (2005), Morley, Nelson and Zivot (2003), Stock and Watson (2007)
- Python Quantlets of this chapter: Quantlets/Ch_06
  - Kalman filter and smoother with exact diffuse initialisation, simulation smoothers, KSC Gibbs sampler, bootstrap and auxiliary particle filters, PMMH and UC-SV written out in `numpy`; checked against `statsmodels`
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source	Use
US GDP price deflator, real GDP (quarterly)	FRED (GDPDEF, GDPC1)	likelihood, MNZ, UC-SV
S&P 500 and BET, daily closes	EODHD	stochastic volatility
HICP of Romania and of the euro area (monthly index, 2025 = 100)	Eurostat prc_hicp_minr	TVP, UC-SV
Romanian real GDP, seasonally and calendar adjusted	Eurostat namq_10_gdp	output gap
US industrial production, payrolls, real income less transfers, real sales (monthly)	FRED (INDPRO, PAYEMS, W875RX1, CMRMTSPL)	DFM

- All sources are public and need no account or key; quarterly series to 2026Q2, monthly series to August or September 2026, daily data to 18 September 2026
- HICP: harmonised index of consumer prices; DFM: dynamic factor model

Origins of the Kalman filter and state space models



Rudolf E. Kálmán, 2007

Photo: ETH-Bibliothek Zürich, Bildarchiv (2007); CC BY-SA 4.0; Wikimedia Commons

- 1960: Kalman (1960) gives the recursive linear filter; it guides the Apollo missions within a decade
- 1982–1997: EM for state space models Shumway and Stoffer (1982); structural time series Harvey (1990); exact diffuse filtering Koopman (1997)
- 1993–1999: particle filters Gordon, Salmond and Smith (1993), Pitt and Shephard (1999); Gibbs sampling of states Carter and Kohn (1994), Frühwirth-Schnatter (1994); stochastic volatility Kim, Shephard and Chib (1998)
- 2002–2010: the simulation smoother Durbin and Koopman (2002); TVP-VAR with stochastic volatility Primiceri (2005), Cogley and Sargent (2005); particle MCMC Andrieu, Doucet and Holenstein (2010)
- Today: central banks track trend inflation and the output gap with these tools Stock and Watson (2007), Morley, Nelson and Zivot (2003)

Prior knowledge from TSA and new material

- Known (TSA, Chapter 10): the state space form, the Kalman filter for the local level by hand, smoothing, ML by the prediction error decomposition, the HP filter as a smoother, Hamilton (2018), Markov switching
- New here
  - the filter and smoother for any linear Gaussian model, with nonstationary states initialised exactly (not with a large variance)
  - inference when a variance may be zero; Bayesian sampling of whole state paths
  - nonlinear and non-Gaussian models: stochastic volatility, particle filters, particle MCMC
  - applications at research level: MNZ, UC-SV, TVP regressions, the ragged edge of a DFM
- Markov switching in depth: Chapter 7; nowcasting with factors: Chapter 5

The general state space form (1/2)

- Two equations: the data are a noisy linear function of an unobserved state, and the state follows a first-order linear dynamic

$$y_t = Z_t\alpha_t + \varepsilon_t, \qquad \alpha_{t+1} = c_t + T_t\alpha_t + R_t\eta_t, \qquad t = 1, \ldots, n$$

- Notation
  - y_t $(p \times 1)$: the observations; α_t $(m \times 1)$: the state; n: the sample length
  - Z_t $(p \times m)$: the measurement matrix; T_t $(m \times m)$: the transition matrix; c_t: a known intercept
  - $\varepsilon_t \sim N(0, H_t)$: measurement noise; $\eta_t \sim N(0, Q_t)$: state shocks; R_t $(m \times r)$ selects the states that receive shocks
  - ε_t and η_s are independent of each other and over time
- Initial state $\alpha_1 \sim N(a_1, P_1)$
  - nonstationary or unknown fixed elements: $P_1 = \kappa P_\infty + P_*$, $\kappa \to \infty$; P_∞ marks the diffuse elements, P_* the covariance of the others

The general state space form (2/2)

- The system matrices Z_t, T_t, R_t, H_t, Q_t depend on parameters ψ; time variation in Z_t gives regressions with time-varying coefficients
  - ARMA, UC, TVP regressions, DFM, VAR with missing data, ETS: all are special cases Durbin and Koopman (2012)
- Notation for the estimates
  - $Y_t = (y_1, \ldots, y_t)$: the data up to t
  - predicted state: $a_t = \mathbb{E}(\alpha_t | Y_{t-1})$, $P_t = \text{Var}(\alpha_t | Y_{t-1})$; filtered: $a_{t|t}$, $P_{t|t}$ (after y_t)
  - smoothed: $\hat{\alpha}_t = \mathbb{E}(\alpha_t | Y_n)$, $V_t = \text{Var}(\alpha_t | Y_n)$ (all the data)

Derivation of the Kalman filter from a Gaussian lemma

- Lemma (conditioning of jointly normal vectors): if (x, y) is Gaussian, $\mathbb{E}(x|y) = \mu_x + \Sigma_{xy}\Sigma_{yy}^{-1}(y - \mu_y)$, $\mathrm{Var}(x|y) = \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx}$
  - μ_x, μ_y: the means; $\Sigma_{xy} = \mathrm{Cov}(x, y)$, $\Sigma_{yy} = \mathrm{Var}(y)$: the covariance blocks
  - apply it to $x = \alpha_t$ and $y = y_t$, conditional on Y_{t-1}: $v_t = y_t - Z_t a_t$ is the new information (the prediction error)
- Prediction-error variance $\mathrm{Var}(v_t|Y_{t-1}) = F_t = Z_t P_t Z_t' + H_t$; $\mathrm{Cov}(\alpha_t, v_t|Y_{t-1}) = P_t Z_t'$; v_t is independent of Y_{t-1}
- Update: $a_{t|t} = a_t + P_t Z_t' F_t^{-1} v_t$, $P_{t|t} = P_t - P_t Z_t' F_t^{-1} Z_t P_t$
- Prediction: $a_{t+1} = c_t + T_t a_{t|t}$, $P_{t+1} = T_t P_{t|t} T_t' + R_t Q_t R_t'$
- Without normality the same recursions give the best *linear* predictor (minimum mean square error among linear functions of Y_t)

The Kalman filter recursions

- With v_t and F_t of the previous slide: $K_t = T_t P_t Z_t' F_t^{-1}$ (Kalman gain), $\quad L_t = T_t - K_t Z_t$
- $a_{t+1} = c_t + T_t a_t + K_t v_t, \quad P_{t+1} = T_t P_t L_t' + R_t Q_t R_t'$
- Cost: $O(n(m^3 + p^3))$; with time-invariant matrices P_t converges to the steady state of the Riccati equation
  - P_t and K_t do not depend on y: they can be computed before the data arrive
- Missing y_t: set $Z_t = 0$ (or drop the missing rows): v_t and K_t vanish and the filter only predicts
  - mixed frequencies and the ragged edge of real-time data are therefore free (Chapter 5)
- Multivariate y_t with diagonal H_t: process the p elements one at a time (univariate treatment Koopman and Durbin (2000)): no $p \times p$ inversion

Smoothing: backward recursions

- State smoother Durbin and Koopman (2012): $r_{t-1} = Z_t' F_t^{-1} v_t + L_t' r_t, \quad N_{t-1} = Z_t' F_t^{-1} Z_t + L_t' N_t L_t, \quad r_n = 0, N_n = 0$
- Smoothed state: $\hat{\alpha}_t = a_t + P_t r_{t-1}, \quad V_t = P_t - P_t N_{t-1} P_t$
  - r_{t-1}: a weighted sum of the innovations from t on; $N_{t-1} = \mathrm{Var}(r_{t-1})$
  - $\hat{\alpha}_t$ corrects a_t by everything learned from t on
- Disturbance smoother: $\hat{\varepsilon}_t = H_t(F_t^{-1} v_t - K_t' r_t)$, $\hat{\eta}_t = Q_t R_t' r_t$: auxiliary residuals for outliers and breaks, and the score of the likelihood
- Derivation from the same lemma, now conditioning on $v_t, \ldots, v_n$, which are independent (Appendix)

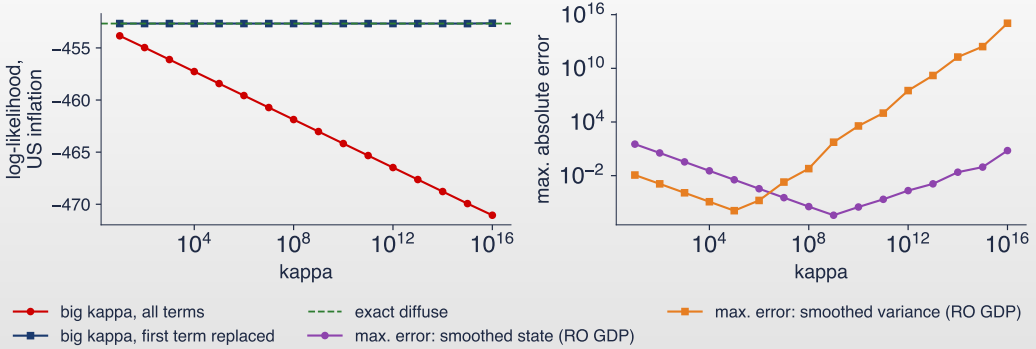
Diffuse initialisation (1/2)

- Random walks, trends and regression coefficients have no stationary distribution: $P_1 = \kappa P_\infty + P_*$, $\kappa \to \infty$
  - "big kappa" ($\kappa = 10^6$–10^7) is an approximation with two failures: the likelihood contains $-\frac{1}{2}\ln\kappa$ terms, and round-off errors grow with κ
- Exact diffuse filter Koopman (1997), Ansley and Kohn (1985), de Jong (1991)
  - expand $P_t = \kappa P_{\infty,t} + P_{*,t} + O(\kappa^{-1})$ and keep both parts while $P_{\infty,t} \neq 0$
  - $P_{\infty,t}$ reaches zero after d steps (d = number of diffuse elements, if they are identified)
  - then the ordinary filter takes over

Diffuse initialisation (2/2)

- The exact diffuse recursions
  - $F_{\infty,t}$, $F_{*,t}$: the diffuse and finite parts of F_t; $K_t^{(\mathbf{0})}$, $K_t^{(\mathbf{1})}$: the matching gains; $L_t^{(\mathbf{0})} = T_t - K_t^{(\mathbf{0})} Z_t$, $L_t^{(\mathbf{1})} = -K_t^{(\mathbf{1})} Z_t$
  - univariate y_t, $F_{\infty,t} = Z_t P_{\infty,t} Z_t' > 0$: $K_t^{(\mathbf{0})} = T_t P_{\infty,t} Z_t' / F_{\infty,t}$, $K_t^{(\mathbf{1})} = T_t (P_{*,t} Z_t' - P_{\infty,t} Z_t' F_{*,t} / F_{\infty,t}) / F_{\infty,t}$
  - $a_{t+\mathbf{1}} = c_t + T_t a_t + K_t^{(\mathbf{0})} v_t$; $P_{\infty,t+\mathbf{1}} = T_t P_{\infty,t} L_t^{(\mathbf{0})\prime}$; $P_{*,t+\mathbf{1}} = T_t P_{\infty,t} L_t^{(\mathbf{1})\prime} + T_t P_{*,t} L_t^{(\mathbf{0})\prime} + R_t Q_t R_t'$
- The smoother has matching diffuse recursions for $r^{(\mathbf{0})}$, $r^{(\mathbf{1})}$, $N^{(\mathbf{0})}$, $N^{(\mathbf{1})}$, $N^{(\mathbf{2})}$ Durbin and Koopman (2012), Section 5.3; in the Quantlet both are checked against `statsmodels`

Exact diffuse against big kappa



⊡ Left: local level for US GDP-deflator inflation at the ML variances; right: local linear trend for 100×log Romanian real GDP, $T = 106$; both against the exact diffuse filter

ATS_ch6_kalman_mle

Interpreting the diffuse initialisation

- With all terms, the big-kappa log-likelihood falls by $\frac{1}{2}\ln 10$ per decade of κ: -460.72 at $\kappa = 10^7$, -468.78 at 10^{14}; the exact value is -452.66
- Replacing the first term by its diffuse limit repairs the level (-452.66 at 10^7), but only because $d = 1$ is known; ML over ψ is unaffected, model comparison across different d is not
- Smoothed variances: error 0.000012 at $\kappa = 10^4$, 0.06 at 10^8 and 10^{15} at 10^{16}: no single κ is safe for both states and variances
- Practical rule: use the exact diffuse filter whenever the model has nonstationary or fixed unknown states

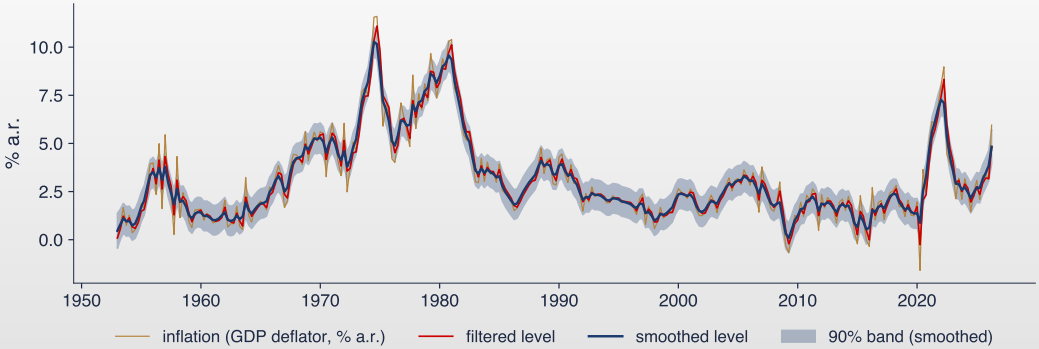
The prediction error decomposition, diffuse case

- $p(y_1, \ldots, y_n) = \prod_t p(y_t|Y_{t-1})$, and $y_t|Y_{t-1} \sim N(Z_t a_t, F_t)$: the likelihood is a by-product of the filter
- Diffuse log-likelihood Durbin and Koopman (2012), eq. (7.4):
  $\ln L_d = -\frac{np}{2} \ln 2\pi - \frac{1}{2} \sum_{t=1}^{d} w_t - \frac{1}{2} \sum_{t=d+1}^{n} (\ln |F_t| + v_t' F_t^{-1} v_t)$
- $w_t = \ln F_{\infty,t}$ if $F_{\infty,t} > 0$, otherwise $w_t = \ln F_{*,t} + v_t^2 / F_{*,t}$: the first d observations only pin down the diffuse states
  - equivalent to the marginal likelihood of de Jong (1991) and to treating the diffuse states as fixed and concentrating them out
- Scale: write $H_t = \sigma^2 H_t^*$, $Q_t = \sigma^2 Q_t^*$; then $F_t = \sigma^2 F_t^*$ and $\hat{\sigma}^2 = \frac{1}{p(n-d)} \sum_{t>d} v_t' F_t^{*-1} v_t$: one parameter fewer in the numerical search

Maximising the likelihood

- Parameterise variances as exp(·), correlations as tanh(·), AR coefficients through partial autocorrelations: unconstrained search, stationary models only
  - several starting values: UC likelihoods are often multimodal (MNZ below)
- Score in one filter-smoother pass: $\partial \ln L/\partial\psi = \frac{1}{2}\sum_t \mathrm{tr}\{(\hat{\varepsilon}_t\hat{\varepsilon}_t' + \mathrm{Var}(\varepsilon_t|Y_n) - H_t)H_t^{-1}\frac{\partial H_t}{\partial\psi}H_t^{-1}\} + \ldots$ Durbin and Koopman (2012), Section 7.3
- EM algorithm Shumway and Stoffer (1982): the E-step is the smoother, the M-step has closed forms for H and Q; slow near the optimum but robust from poor starts (large DFM)
- Standard errors: inverse Hessian at $\hat{\psi}$ (numerical); valid only for interior points; report them on the transformed scale
- Identification: some UC models are observationally equivalent to ARIMA models with fewer parameters; check the reduced form before interpreting the components

Local level for US inflation: numpy against statsmodels



- GDP-deflator inflation, 1953Q1–2026Q2, $T = 294$; exact diffuse ML; the smoothed level with a 90% band and the one-sided (filtered) level

ATS_ch6_kalman_mle

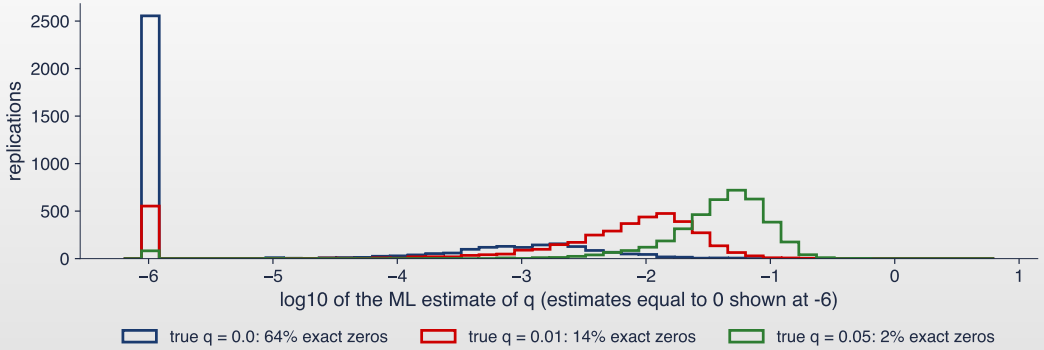
Interpreting the local level estimates

- numpy: $\hat{\sigma}_\varepsilon^2 = 0.517$, $\hat{\sigma}_\eta^2 = 0.453$, $\hat{q} = 0.877$; statsmodels: 0.517 and 0.453: the same optimum
- Log-likelihoods -452.66 and -451.74: they differ by exactly $\frac{1}{2}\ln 2\pi = 0.919$, the constant that `statsmodels` omits for the $d = 1$ diffuse observation
- Steady-state gain 0.596: each new quarter moves the level estimate by 60% of the surprise; inflation is closer to a random walk than to white noise around a mean
- Last smoothed level 4.83% (s.d. 0.55): at the end of the sample the smoothed and filtered estimates coincide, and so does their uncertainty
- One constant variance for 1953–2026 is implausible (Great Inflation, Great Moderation): this motivates UC-SV in the last section

Inference at the boundary: the pile-up problem

- Local level with $q = \sigma_\eta^2/\sigma_\varepsilon^2$: the ML estimate equals exactly zero with positive probability even when $q > 0$ Shephard and Harvey (1990)
  - the score at $q = 0$ is often negative because a smooth trend fits the data better than a slightly wandering one
- Consequences
  - the confidence interval computed from the Hessian is invalid
  - the LR test of $q = 0$ is not χ^2_1, not even $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$: the model under the alternative is nonstationary
  - remedies: median-unbiased estimation of a coefficient variance Stock and Watson (1998); parametric bootstrap of the LR statistic (TVP section); Bayesian priors that keep q away from zero
- Same issue in TVP regressions (is the coefficient constant?) and in UC models (is the trend deterministic?)

Frequency of a zero estimate of the level variance



- 4 000 simulated local level series of length $T = 100$ for each true q; ML of q on a fine grid with σ_ε^2 concentrated out

ATS_ch6_kalman_mle

Interpreting the pile-up simulation

- True $q = 0$: 64% of the estimates are exactly zero; the rest spread over several orders of magnitude
- True $q = 0.01$: still 14% zeros; median estimate 0.0077; with $q = 0.05$: 2% zeros
- A reported $\hat{\sigma}_\eta^2 = 0$ is therefore not evidence of a constant level; a small positive value is compatible with a deterministic one
- Report the profile likelihood or the posterior of q, not a point estimate with a Hessian standard error

Sampling the states



Portrait traditionally identified as Thomas Bayes (identification uncertain)

Portrait: unknown author; public domain; Wikimedia Commons

- Target: $p(\alpha_{\mathbf{1}}, \ldots, \alpha_n, \psi | Y_n)$; the filter gives $p(\alpha_t | Y_t, \psi)$ for one t and one ψ only
- Gibbs sampler: alternate $\alpha_{\mathbf{1}:n} \sim p(\alpha_{\mathbf{1}:n} | \psi, Y_n)$ and $\psi \sim p(\psi | \alpha_{\mathbf{1}:n}, Y_n)$; given the states, ψ is often a conjugate regression problem
- Uncertainty about ψ enters the bands of the states (ML bands ignore it)
- Functions of whole paths become easy: the date of a peak, the probability that a trend stays inside a target band
- Sampling the whole path in one block (not α_t one at a time) is what makes the sampler mix

Forward filtering, backward sampling

- Factorisation: $p(\alpha_{1:n}|Y_n) = p(\alpha_n|Y_n)\prod_{t=1}^{n-1} p(\alpha_t|\alpha_{t+1}, Y_t)$ (Markov property of the state)
- Forward: run the filter, store $a_{t|t}, P_{t|t}$; draw $\alpha_n \sim N(a_{n|n}, P_{n|n})$ Carter and Kohn (1994), Frühwirth-Schnatter (1994)
- Backward: $\alpha_t|\alpha_{t+1}, Y_t \sim N(m_t, S_t)$ by the lemma, with the smoothing gain $G_t = P_{t|t}T_t'P_{t+1}^{-1}$ (m_t, S_t: conditional mean and variance):
  - $m_t = a_{t|t} + G_t(\alpha_{t+1} - c_t - T_t a_{t|t}), \quad S_t = P_{t|t} - G_t T_t P_{t|t}$
- Issues: P_{t+1} is singular when $R_t Q_t R_t'$ is (lags in the state): use the rows with shocks only; n draws of dimension m
- Same output as the Gibbs step in the Bayesian VAR and DFM samplers of Chapter 5

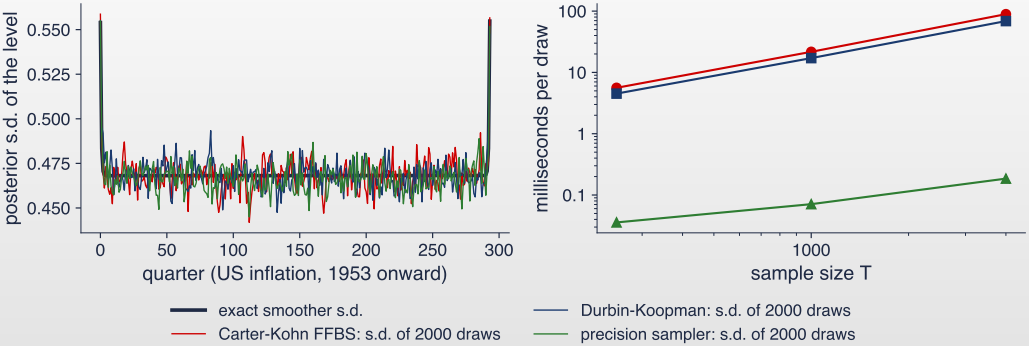
The Durbin–Koopman simulation smoother

- Idea Durbin and Koopman (2002): $\alpha - \hat{\alpha}(y)$ has the same conditional distribution for every y (Gaussian model): simulate it once from the model
- Algorithm (mean correction):
  - 1. draw $\alpha^+_{1:n}, y^+_{1:n}$ from the model (shocks and initial state); keep the missing pattern of y
  - 2. run one state smoother on $y - y^+$ (zero intercepts): $\hat{\alpha}(y - y^+) = \hat{\alpha}(y) - \hat{\alpha}(y^+)$
  - 3. $\tilde{\alpha} = \alpha^+ + \hat{\alpha}(y) - \hat{\alpha}(y^+)$ is a draw from $p(\alpha|Y_n)$
- Only the mean smoother is needed: no $P_{t|t}$ inversions, no singularity problems, diffuse states handled by the exact diffuse smoother (proof of exactness: Appendix)
- Precursor: the disturbance simulation smoother of de Jong and Shephard (1995); same idea for the disturbances ε_t, η_t

The precision sampler

- Stack the model Chan and Jeliazkov (2009): $y = X\alpha + \varepsilon$, $D\alpha = \eta$
  - D: a banded difference matrix (first differences for a random walk)
  - then $\alpha | y \sim N(\hat{\alpha}, K^{-1})$ with $K = D'\Sigma_\eta^{-1} D + X'\Sigma_\varepsilon^{-1} X$
- K is banded (tridiagonal for a random walk): one banded Cholesky factor $K = U'U$ costs $O(n)$
  - α, y: all states and observations stacked; X: block-diagonal of the Z_t; Σ_ε, Σ_η: the stacked noise covariances; K: the posterior precision (inverse covariance)
  - $\hat{\alpha}$ solves $K\hat{\alpha} = X'\Sigma_\varepsilon^{-1} y$; a draw is $\hat{\alpha} + U^{-1} z$, $z \sim N(0, I)$
- Time-varying variances (stochastic volatility) only change the diagonal weights: ideal inside Gibbs samplers for UC-SV and SV
- Three samplers, one target: FFBS, DK and precision sampling draw from the same $p(\alpha | Y_n, \psi)$; they differ in cost and generality

Comparison of the three state samplers



- Local level for US inflation at the ML variances, proper prior $\mu_1 \sim N(y_1, 100)$; left: s.d. of 2 000 draws against the exact smoother; right: time per draw for simulated samples

ATS_ch6_simulation_smoother

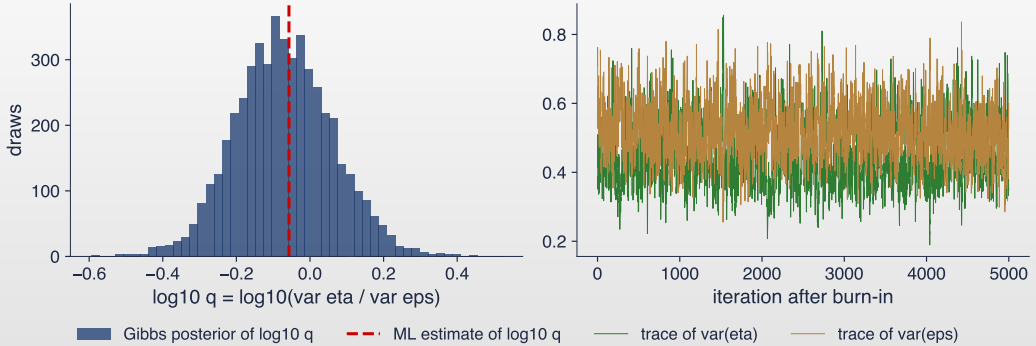
Interpreting the three samplers

- Ratios of draw s.d. to exact s.d. lie in [0.94; 1.05] (FFBS), [0.96; 1.05] (DK) and [0.95; 1.04] (precision): Monte Carlo noise only
- Largest error of the draw means: 0.031, 0.027, 0.039, of the order of the Monte Carlo standard error
- For $T = 4000$: 89.2 ms (FFBS), 68.8 ms (DK), 0.19 ms (precision) per draw; all linear in T, the banded solver has no Python loop
- Choose DK for general models with diffuse states, precision sampling for random walks with time-varying variances

A Gibbs sampler for the local level model

- Model: $y_t = \mu_t + \varepsilon_t$, $\mu_{t+1} = \mu_t + \eta_t$; μ_t: the level; $IG(a_0, b_0)$: inverse gamma prior with shape a_0 and scale b_0; $q = \sigma_\eta^2/\sigma_\varepsilon^2$
- Block 1: $\mu_{1:n}|\sigma_\varepsilon^2, \sigma_\eta^2, Y_n$ by the DK simulation smoother (exact diffuse μ_1)
- Block 2: $\sigma_\varepsilon^2|\mu, Y_n \sim IG(a_0 + \frac{n}{2}, b_0 + \frac{1}{2}\sum_t (y_t - \mu_t)^2)$; $\sigma_\eta^2|\mu \sim IG(a_0 + \frac{n-1}{2}, b_0 + \frac{1}{2}\sum_t (\Delta\mu_t)^2)$
- Prior $IG(2.5, 0.25)$ for both: proper, weak, and it rules out $\sigma_\eta^2 = 0$ exactly
  - the prior is not innocent near the boundary: a conjugate IG pushes small variances up; the non-centred parameterisation of Frühwirth-Schnatter and Wagner (2010) allows a normal prior on $\pm\sigma_\eta$ instead
- Convergence: trace plots, several chains, inefficiency factor $1 + 2\sum_k \rho_k$ (Parzen weights): the number of draws worth one independent draw Kim, Shephard and Chib (1998)

Gibbs posterior of the signal-to-noise ratio



⊡ US inflation, local level, 5 000 draws after burn-in; left: posterior of $\log_{10} q$ and the ML value; right: traces of the two variances

ATS_ch6_simulation_smoother

Interpreting the Gibbs output

- Posterior median of q: 0.84, 90% credible interval [0.52; 1.42]; ML: 0.88
- Posterior means 0.515 and 0.441 against ML 0.517 and 0.453: with $n = 294$ the prior barely matters away from the boundary
- Inefficiency factors 9.9 and 11.3: the variance draws inherit the autocorrelation of the state draws
- The interval for q excludes zero: the trend of US inflation moves, but the upper end of the interval is 2.7 times the lower end

The stochastic volatility model



Neil Shephard, 2004

Photo: Renate Schmid (2004), MFO; CC BY-SA 2.0 de; Wikimedia Commons

- $y_t = \exp(h_t/2)\,\epsilon_t, \quad h_{t+1} = \mu + \phi(h_t - \mu) + \sigma_\eta \eta_t, \quad \epsilon_t, \eta_t \sim$ i.i.d. $N(0,1)$ Taylor (1994)
- h_t: log-variance, a latent AR(1) state; ϕ: persistence; σ_η: volatility of volatility; $\exp(\mu/2)$: typical volatility
- A state space model with a nonlinear measurement equation: the Kalman filter does not apply directly
- Continuous-time counterpart: the log-variance as an Ornstein–Uhlenbeck process (option pricing, realised measures in Chapter 8)

Stochastic volatility and GARCH compared (1/2)

- GARCH Bollerslev (1986): $\sigma_t^2 = \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2$, $\omega > 0$, $\alpha, \beta \geq 0$
  - σ_t^2 is known at $t-1$ (a function of past returns): one source of randomness
  - the likelihood is a product of known densities: one pass, exact ML
- SV: h_t has its own shock η_t; even with all past returns, today's variance is uncertain
  - $p(y_{1:n}|\psi) = \int \prod_t N(y_t; 0, e^{h_t})\, p(h_{1:n}|\psi)\, dh_{1:n}$: an n-dimensional integral without closed form
- Comparison on the likelihood scale is possible only after the integral is computed: particle filter (next section) or importance sampling Kim, Shephard and Chib (1998)

Stochastic volatility and GARCH compared (2/2)

- Moments, with $\sigma_h^2 = \sigma_\eta^2/(1-\phi^2)$ the unconditional variance of h_t
  - kurtosis of y_t: $3\exp(\sigma_h^2) > 3$ even with Gaussian ϵ_t: fat tails come from the random variance
  - $\text{Corr}(y_t^2, y_{t-k}^2) = (e^{\sigma_h^2\phi^k} - 1)/(3e^{\sigma_h^2} - 1)$: volatility clustering that decays with ϕ^k
- Leverage enters as $\text{Corr}(\epsilon_t, \eta_t) = \rho < 0$ Omori, Chib, Shephard and Nakajima (2007); GARCH needs an asymmetric term for the same effect

A linear state space form: log-squared returns

- $y_t^* = \ln y_t^2 = h_t + \xi_t$, $\xi_t = \ln \epsilon_t^2$: linear in h_t, but ξ_t follows a $\ln \chi_1^2$ law with mean -1.2704 and variance $\pi^2/2 \approx 4.93$
- QML Harvey, Ruiz and Shephard (1994): run the Kalman filter as if ξ_t were $N(-1.2704, \pi^2/2)$; consistent, but inefficient
  - y_t^* is ARMA(1,1): $\text{Corr}(y_t^*, y_{t-k}^*) = \phi^k \sigma_h^2/(\sigma_h^2 + \pi^2/2)$, small because the noise dominates the signal
- Zero returns: $\ln 0 = -\infty$; use $y_t^* = \ln(y_t^2 + c)$ with a small offset c (here $c = 10^{-3}\text{Var}(y)$, scaled to the series)
- Multivariate extension of the same idea: factor SV models Harvey, Ruiz and Shephard (1994)

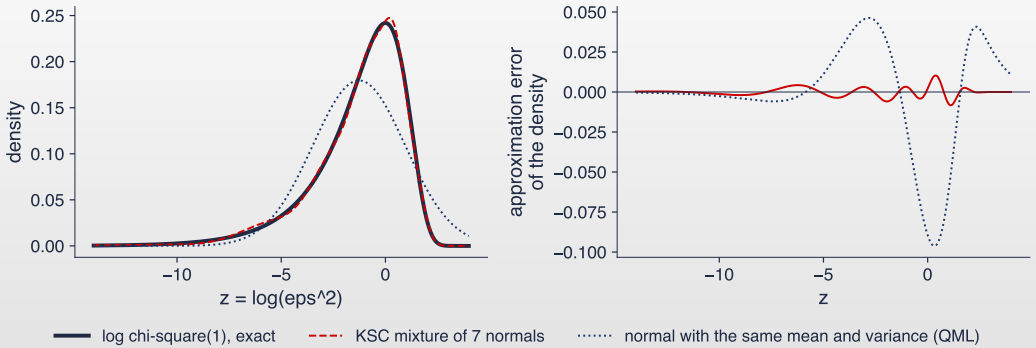
The KSC mixture: a conditionally Gaussian model

- Kim, Shephard and Chib (1998): approximate $\ln \chi^2_1$ by a mixture of seven normals; given the component s_t, the model is linear and Gaussian
- $y_t^* = h_t + m_{s_t} - 1.2704 + \sqrt{v_{s_t}}\, u_t$, $u_t \sim N(0,1)$, $P(s_t = i) = p_i$; $s_t \in \{1, \ldots, 7\}$: the component; p_i, m_i, v_i: its probability, mean and variance (table)
- Mixture mean -1.2704 and variance 4.935, against -1.2704 and 4.935 for the exact law
- The residual approximation error can be removed by reweighting the draws (importance weights) Kim, Shephard and Chib (1998); finer 10-component mixtures Omori, Chib, Shephard and Nakajima (2007)

i	p_i	m_i	v_i
1	0.00730	−10.12999	5.79596
2	0.10556	−3.97281	2.61369
3	0.00002	−8.56686	5.17950
4	0.04395	2.77786	0.16735
5	0.34001	0.61942	0.64009
6	0.24566	1.79518	0.34023
7	0.25750	−1.08819	1.26261

- Source: Kim, Shephard and Chib (1998), Table 4

The log chi-square law and its approximations



- Left: density of $\ln \epsilon^2$ (exact), the KSC mixture and the normal with the same two moments used by QML; right: the two approximation errors

ATS_ch6_stochastic_volatility

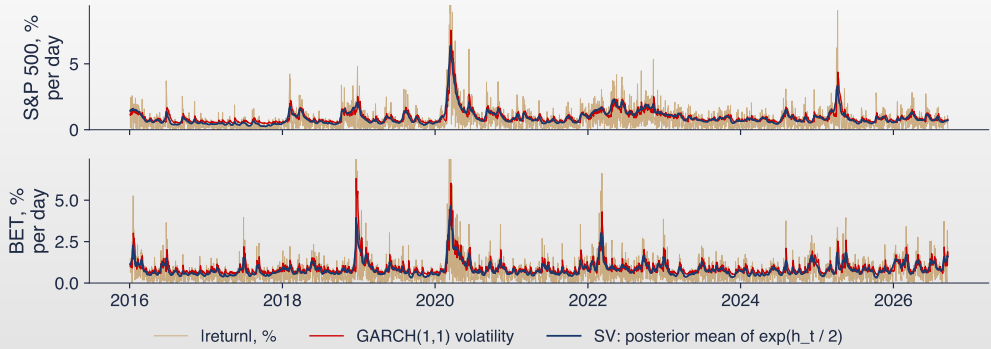
Interpreting the mixture approximation

- The law is strongly skewed to the left: small returns produce very negative $\ln y_t^2$; the normal approximation misses both the skew and the peak
- Largest density error: 0.010 for the mixture, 0.096 for the normal (QML)
- QML treats a skewed noise as symmetric: the filter overreacts to near-zero returns; the mixture fixes this at the cost of one extra Gibbs block

Gibbs sampling of the SV model

- Block 1: $s_t | y_t^*, h_t$ independently, $P(s_t = i) \propto p_i \, N(y_t^* - h_t;\ m_i - 1.2704, v_i)$
- Block 2: $h_{\mathbf{1}:n} | \mathbf{s}, \psi, y^*$ in one draw: a linear Gaussian model with known time-varying measurement variances v_{s_t} (precision sampler or DK)
- Block 3: $\sigma_\eta^2 | h, \mu, \phi$ inverse gamma; ϕ by Metropolis–Hastings with the AR(1) regression as proposal; μ normal
- Priors Kim, Shephard and Chib (1998): $(\phi + 1)/2 \sim \mathrm{Beta}(20, 1.5)$, $\sigma_\eta^2 \sim IG(2.5, 0.025)$, $\mu \sim N(0, 10)$
- Single-move samplers (one h_t at a time, Jacquier, Polson and Rossi (1994)) mix very slowly because consecutive h_t are highly correlated: sample the path as a block

Stochastic volatility of the S&P 500 and the BET



- Daily log returns in %, demeaned, 2016–2026 ($T = 2692$ and 2675); KSC Gibbs sampler, 20 000 draws; GARCH(1,1) by ML on the same data

ATS_ch6_stochastic_volatility

Interpreting the SV estimates

Index	ϕ	σ_η	μ	ineff. ϕ, σ_η	GARCH $\alpha + \beta$	corr. vol.
S&P 500	0.963	0.289	−0.465	49 / 98	0.967	0.91
BET	0.914	0.393	−0.624	58 / 87	0.952	0.84

- 90% intervals for ϕ: [0.950; 0.975] (S&P 500), [0.888; 0.938] (BET): very persistent log-variance, as in Kim, Shephard and Chib (1998)
- The two volatility paths are close (correlation in the last column), but SV reacts less to a single large return than GARCH, whose variance jumps by $\alpha y_{t-\mathbf{1}}^{\mathbf{2}}$
- Large inefficiency factors for σ_η are typical: σ_η and the path $h_{\mathbf{1}:n}$ are strongly dependent; long chains or interweaving are needed
- Kurtosis of returns: 19.5 (S&P 500) and 20.8 (BET); with Gaussian ϵ_t, SV explains only part of it (SV-t errors are the usual extension)

The Bayes filter

- General model: $y_t \sim g(y_t|x_t, \psi)$, $x_t \sim f(x_t|x_{t-1}, \psi)$; g: measurement density, f: transition density, x_t: the state; target $p(x_t|Y_t)$ and $p(y_t|Y_{t-1})$
- Prediction: $p(x_t|Y_{t-1}) = \int f(x_t|x_{t-1})\, p(x_{t-1}|Y_{t-1})\, dx_{t-1}$
- Update: $p(x_t|Y_t) = g(y_t|x_t)\, p(x_t|Y_{t-1})/p(y_t|Y_{t-1}), \quad p(y_t|Y_{t-1}) = \int g(y_t|x_t)\, p(x_t|Y_{t-1})\, dx_t$
- Closed form only in two cases: linear Gaussian (Kalman) and finite state spaces (the Hamilton filter of Markov switching, Chapter 7)
  - otherwise: approximate the model (EKF), approximate the densities by points (UKF), or by weighted samples (particle filters)

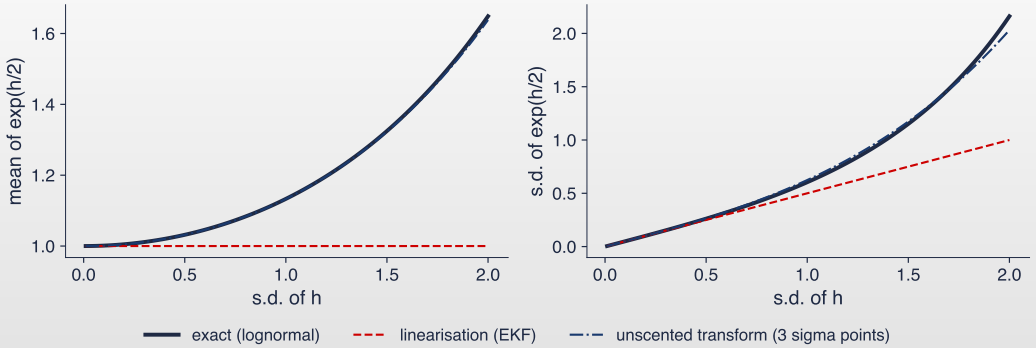
The extended Kalman filter

- Nonlinear model $y_t = Z(x_t) + \varepsilon_t$, $x_{t+1} = T(x_t) + R\eta_t$: linearise around the current estimates, $\dot{Z}_t = \partial Z/\partial x\,|_{a_t}$, $\dot{T}_t = \partial T/\partial x\,|_{a_{t|t}}$
- Run the Kalman recursions with $\dot{Z}_t$, $\dot{T}_t$ for the variances and the exact functions for the means Durbin and Koopman (2012), Section 10.2
- A failure that teaches: in SV, $\mathbb{E}(y_t|h_t) = 0$ for every h_t, so $\partial\mathbb{E}(y_t|h_t)/\partial h_t = 0$: the EKF gain is zero and the filter never learns h_t
  - the information about h_t is in y_t^2, a second moment; linearisation keeps only first moments
- Rule: transform the model first ($\ln y_t^2$), or use a method that propagates the whole distribution

The unscented Kalman filter

- Unscented transform Julier and Uhlmann (2004): $N(m, P)$ in n dimensions is represented by $2n+1$ sigma points
  - $\mathcal{X}_0 = m$, $\mathcal{X}_{\pm i} = m \pm (\sqrt{(n+\lambda)P})_i$; $(\sqrt{A})_i$: the i-th column of a matrix square root
  - weights $w_0 = \lambda/(n+\lambda)$, $w_{\pm i} = 1/(2(n+\lambda))$; λ: a spread parameter
  - push each point through g and take weighted means and covariances
- Exact for the mean to second order of the Taylor expansion of g (the EKF: first order); no Jacobians needed
- UKF: apply the transform in the prediction and in the update step; cost similar to the EKF
- Still Gaussian at each step: multimodal or heavy-tailed filtering densities need particles

Linearisation against the unscented transform



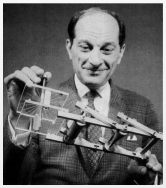
⊡ Volatility from log-variance: mean and s.d. of $\exp(h/2)$ for $h \sim N(0, s^2)$; exact lognormal moments, first-order linearisation (EKF) and the unscented transform with three sigma points

ATS_ch6_nonlinear_filters

Interpreting the unscented transform

- At $s = 1$: exact mean 1.131, linearisation 1.000, unscented 1.131; exact s.d. 0.598, linearisation 0.496, unscented 0.618
- Linearisation ignores the convexity of exp: it underestimates expected volatility, more so when h is uncertain
- Three sigma points capture the convexity almost exactly for the mean; the variance is off for large s
- Same bias in practice: an EKF forecast of volatility or of an option price is too low when the state is uncertain

Particle filters



Stanisław Ulam with the FERMIAC, a Monte Carlo device

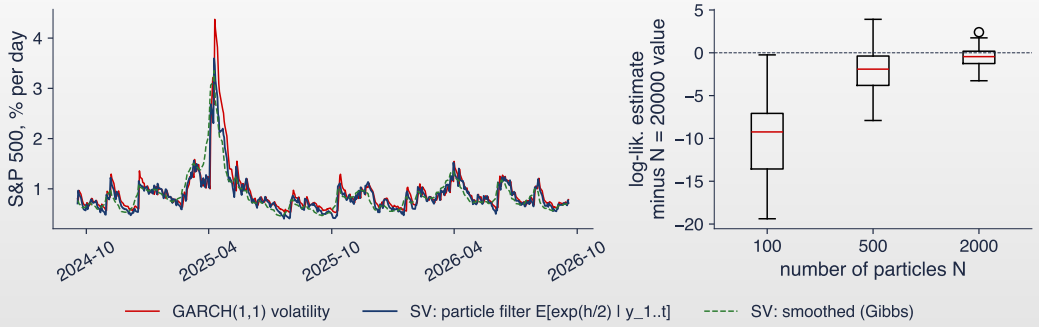
Photo: Los Alamos National Laboratory (LA-UR-00-2532); public domain; Wikimedia Commons

- Represent $p(x_t|Y_t)$ by particles $x_t^{(i)}$ with weights $W_t^{(i)}$, $i = 1, \ldots, N$; Monte Carlo goes back to Ulam and Metropolis (1949)
- Bootstrap filter Gordon, Salmond and Smith (1993), Kitagawa (1996)
  - propagate: $x_t^{(i)} \sim f(\cdot|x_{t-\mathbf{1}}^{(i)})$
  - weight: $w_t^{(i)} = g(y_t|x_t^{(i)})$
  - resample N particles with probabilities $W_t^{(i)} \propto w_t^{(i)}$
- Without resampling the weights degenerate (one particle takes all the weight); effective sample size $\mathrm{ESS}_t = 1/\sum_i (W_t^{(i)})^{\mathbf{2}}$
- Systematic resampling: one uniform U, points $(U + i - 1)/N$ on the cumulative weights: lower variance than multinomial

The particle likelihood

- $\hat{p}(y_t|Y_{t-1}) = \frac{1}{N}\sum_i w_t^{(i)}$ and $\hat{L}(\psi) = \prod_t \hat{p}(y_t|Y_{t-1})$
- $\mathbb{E}\hat{L}(\psi) = L(\psi)$ exactly, for any $N \geq 1$ Del Moral (2004) (Appendix)
  - $\hat{L}$ is unbiased on the level scale
  - not on the log scale: if $\ln \hat{L} \approx N(\ln L - \tau^2/2, \tau^2)$, then $\mathbb{E}\ln \hat{L} = \ln L - \tau^2/2$; $\tau^2 = \mathrm{Var}\ln \hat{L}$
- $\mathrm{Var}\ln \hat{L}$ grows roughly linearly in n and falls like $1/N$: double the sample, double the particles
- $\hat{L}(\psi)$ is not smooth in ψ (resampling is discontinuous): do not maximise it by gradient methods; use it inside MCMC

Particle filter for the S&P 500 SV model



- Left: filtered volatility $\mathbb{E}[\exp(h_t/2)|Y_t]$ (bootstrap filter, $N = 20\,000$, Gibbs posterior mean of ψ) against GARCH(1,1) and the smoothed SV path, last two years; right: log-likelihood estimates, 100 runs per N

ATS_ch6_stochastic_volatility

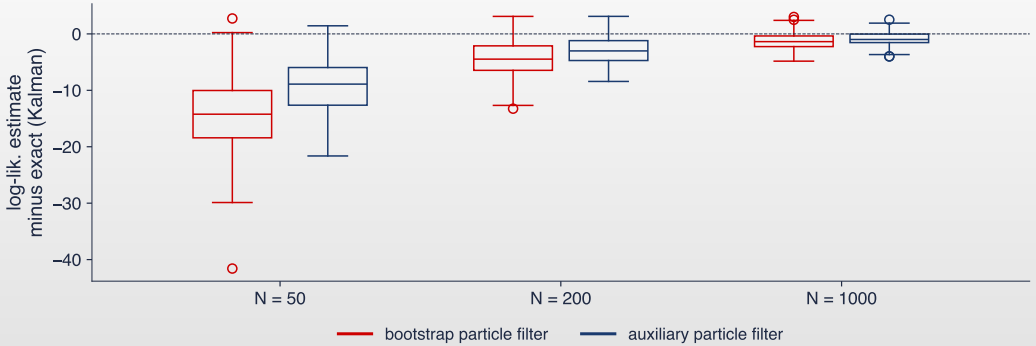
Interpreting the particle filter

- Monte Carlo s.d. of $\ln \hat{L}$: 4.19 ($N = 100$), 2.38 ($N = 500$), 1.14 ($N = 2000$); small N also biases $\ln \hat{L}$ downwards
- SV log-likelihood -3378.8 against the GARCH(1,1) maximum -3467.3: a difference of 88.5 with the same number of parameters
- The comparison favours SV even at the posterior mean (not the maximum) of its likelihood; Kim, Shephard and Chib (1998) compare SV with ARCH models on the same likelihood scale
- The filtered SV volatility is a one-sided estimate, like GARCH: correlation 0.96; the smoothed path anticipates turning points because it uses future data
- Effective sample size over the sample: minimum 150, median 18719 of 20 000 particles

The auxiliary particle filter

- Bootstrap weakness: particles are propagated blind to y_t; an outlier leaves few particles with weight
- APF Pitt and Shephard (1999): look ahead before propagating
  - first stage: resample $x_{t-1}^{(i)}$ with weights $\propto W_{t-1}^{(i)}\, g(y_t|\mu_t^{(i)})$, $\mu_t^{(i)} = \mathbb{E}(x_t|x_{t-1}^{(i)})$
  - second stage: propagate, then weight by $g(y_t|x_t^{(i)})/g(y_t|\mu_t^{(k_i)})$ to correct the look-ahead
  - k_i: the index of the particle selected in the first stage
- Fully adapted case: first-stage weights $p(y_t|x_{t-1})$ and proposal $p(x_t|x_{t-1}, y_t)$: second-stage weights are equal (locally optimal)
- The likelihood estimate stays unbiased: $\hat{p}(y_t|Y_{t-1}) = \left(\sum_i W_{t-1}^{(i)} g(y_t|\mu_t^{(i)})\right) \cdot \frac{1}{N}\sum_i \omega_t^{(i)}$; $\omega_t^{(i)}$: the second-stage weights

Particle likelihoods against the exact Kalman likelihood



- Local level model for US inflation at the ML variances: error of $\ln \hat{L}$ (given y_1) for the bootstrap and the auxiliary filter, 200 runs per N

ATS_ch6_nonlinear_filters

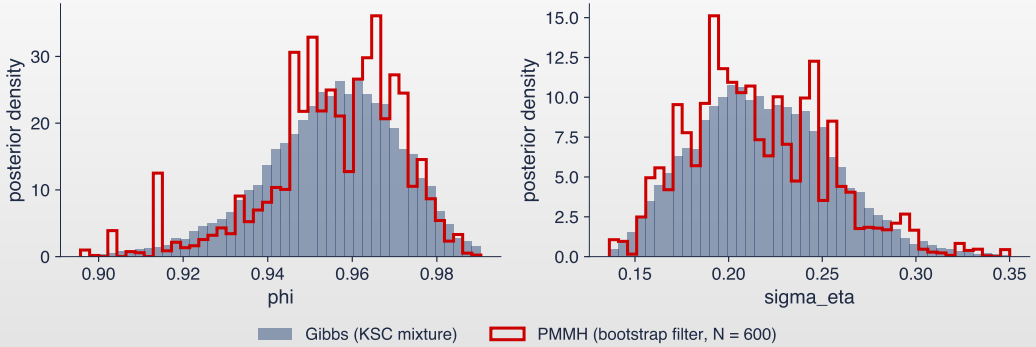
Interpreting the particle check

- Mean error of $\ln \hat{L}$: -14.37, -4.48, -1.30 (bootstrap, $N = 50, 200, 1000$) and -9.37, -2.87, -0.88 (auxiliary)
- Standard deviations: 1.46 (bootstrap) and 1.17 (auxiliary) at $N = 1000$: the look-ahead buys precision because the signal-to-noise ratio is high
- The negative mean is the Jensen bias of a log of an unbiased estimator: it shrinks like $\mathrm{Var}(\ln \hat{L})/2$
- Always test a particle filter on a model with a known likelihood before using it where none is available

Particle MCMC

- Pseudo-marginal idea Andrieu and Roberts (2009): run Metropolis–Hastings with an unbiased, positive estimate $\hat{L}(\psi)$ in place of $L(\psi)$; the chain still targets the exact posterior $p(\psi|Y_n)$
- PMMH Andrieu, Doucet and Holenstein (2010)
  - propose ψ' from a proposal density $q(\cdot|\psi)$; run a particle filter for $\hat{L}(\psi')$
  - accept with probability $\min\{1, \hat{L}(\psi')p(\psi')q(\psi|\psi')/[\hat{L}(\psi)p(\psi)q(\psi'|\psi)]\}$; $p(\cdot)$: the prior
  - keep $\hat{L}$ of the current point (do not recompute it)
- Choosing N Pitt, Silva, Giordani and Kohn (2012), Doucet, Pitt, Deligiannidis and Kohn (2015)
  - too few particles make the chain stick; too many waste time
  - target: $\mathrm{Var}(\ln \hat{L}) \approx 1$–$2$ at the posterior mean
  - particle Gibbs (conditional SMC) samples the state path too; useful when no conditionally Gaussian structure exists
- Works for any model you can simulate and whose measurement density you can evaluate: SV with jumps, DSGE models, nonlinear UC models

PMMH against the KSC Gibbs sampler



- Last 1000 S&P 500 returns; PMMH with a bootstrap filter of $N = 600$ particles, 10 000 iterations, random walk on $(\mu, \operatorname{atanh} \phi, \ln \sigma_\eta)$; Gibbs with the KSC mixture

ATS_ch6_stochastic_volatility

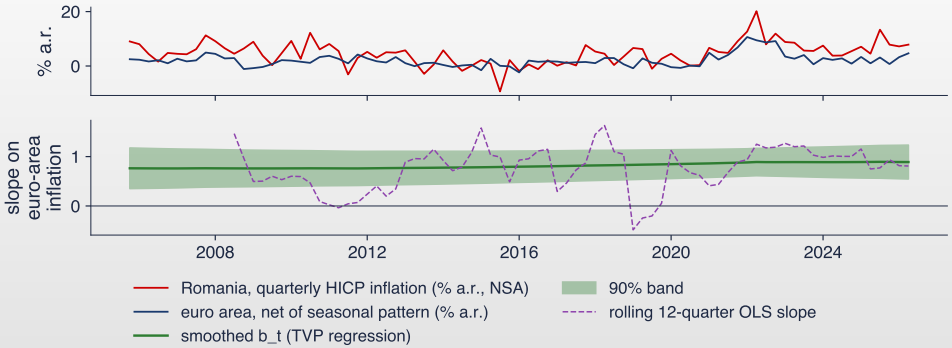
Interpreting the PMMH run

- Posterior means of ϕ: 0.955 (Gibbs) and 0.955 (PMMH); of σ_η: 0.219 and 0.216; posterior s.d. 0.037 and 0.039
- Two exact algorithms, one posterior: the KSC mixture approximation has no visible effect here
- $N = 600$ gives s.d. of $\ln \hat{L}$ equal to 1.19 at the posterior mean, inside the recommended range; acceptance rate 13%
- Cost: 3.6 minutes for PMMH against seconds for Gibbs: use PMMH when there is no conditionally Gaussian representation

A TVP regression in state space form

- $y_t = c_t + b_t x_t + D_t'\gamma + \varepsilon_t, \quad c_{t+1} = c_t + \eta_{c,t}, \quad b_{t+1} = b_t + \eta_{b,t}$
  - c_t, b_t: intercept and slope, random walks with variances σ_c^2, σ_b^2; D_t: dummies with fixed γ
  - state $\alpha_t = (c_t, b_t, \gamma')'$; $Z_t = (1, x_t, D_t')$
- Fixed coefficients (γ, here quarterly seasonal dummies) are states with zero variance and a diffuse start: $d =$ number of states $= 5$
  - with $\sigma_c^2 = \sigma_b^2 = 0$ the smoother returns the OLS estimates (recursive least squares)
- Question: has the co-movement of Romanian inflation with euro-area inflation changed since inflation targeting began (2005)?
- y_t: Romanian HICP inflation, quarterly, % a.r., not seasonally adjusted; x_t: euro-area HICP inflation net of its seasonal pattern; 2005Q4–2026Q2, $T = 83$
- Test of a constant slope: H_0: $\sigma_b^2 = 0$ on the boundary; parametric bootstrap of the LR statistic from the restricted model (199 samples)

A time-varying slope of Romanian on euro-area inflation



- Top: the two inflation rates; bottom: smoothed b_t with a 90% band (exact diffuse ML) and the slope of a rolling 12-quarter regression with the same dummies

ATS_ch6_tvp_inflation

Interpreting the TVP regression

- ML: $\hat{\sigma}_c^2 = 0.332$, $\hat{\sigma}_b^2 = 0.0011$, $\hat{\sigma}_\varepsilon^2 = 7.21$; numpy and `statsmodels` log-likelihoods -209.93 and -209.93
- b_t drifts from 0.76 (s.e. 0.23) in 2008Q3 to 0.89 (s.e. 0.17) in 2022Q4: a slope close to one, but no significant change
- LR $= 0.066$; bootstrap $p = 0.265$ (95% quantile 1.91; 66% of bootstrap statistics are exactly zero: pile-up again); the $\frac{1}{2}\chi^2_1$ rule gives $p = 0.399$
- The rolling regression swings between negative values and 1.5: that is sampling noise of 12-quarter windows, not time variation; the state space model averages it out with an explicit variance

TVP-VAR with stochastic volatility



Thomas J. Sargent, 2011

Photo: Holger Motzkau (2011); CC BY-SA 3.0; Wikimedia Commons

- Cogley and Sargent (2001), Cogley and Sargent (2005), Primiceri (2005): $y_t = X_t'\beta_t + A_t^{-1}\Sigma_t\epsilon_t$
  - X_t: the lags; β_t: the coefficients; $\epsilon_t \sim N(0, I)$
  - A_t: lower triangular with free elements a_t; $\Sigma_t = \mathrm{diag}(\sigma_{j,t})$
  - β_t, a_t and $\ln \sigma_{j,t}$ follow random walks
- Question: did US monetary policy change (drifting coefficients) or did the shocks shrink (falling volatilities) after 1980?
- Answer of Primiceri (2005): the systematic response of policy did change, but these changes explain little of the 1970s; the variance of non-policy shocks matters more
- Without stochastic volatility, falling shock variances are wrongly attributed to changing coefficients

Estimating a TVP-VAR-SV

- Gibbs blocks, each one of the samplers of this chapter:
  - $\beta_{1:n} \mid$ rest: simulation smoother of a linear Gaussian model with known time-varying covariances
  - $a_{1:n} \mid$ rest: equation by equation, the same smoother
  - $\ln \sigma_{j,1:n} \mid$ rest: the KSC mixture with its indicators
  - covariances of the random walks: inverse Wishart
- The order of the Gibbs steps matters: the original algorithm drew the mixture indicators at the wrong point of the cycle; Del Negro and Primiceri (2015) give the corrected order, and the substantive results change little
- Priors from a training sample at the start of the data, as in Primiceri (2005), with small scales for the random-walk covariances: the prior decides how much variation is allowed
- Overparameterisation: with k coefficients per equation, shrink the variances, or select which coefficients vary (non-centred parameterisation, Frühwirth-Schnatter and Wagner (2010))

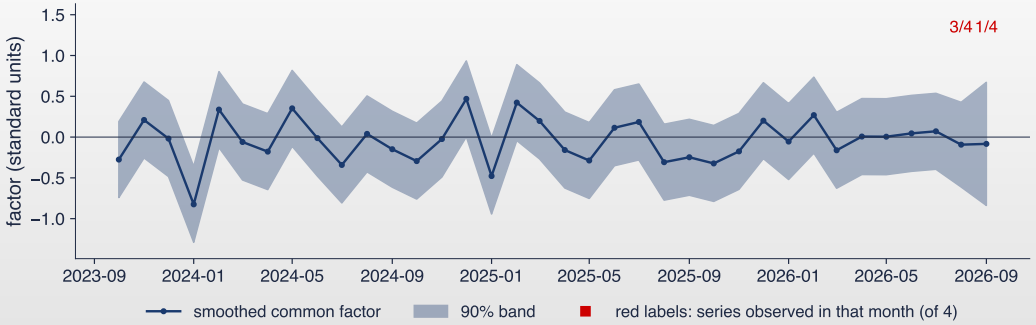
The DFM as a state space model (1/2)

- $x_{it} = \lambda_i' f_t + e_{it}, \quad f_t = \Phi f_{t-1} + u_t, \quad e_{it} = \rho_i e_{i,t-1} + \nu_{it}$
  - f_t: the factors, a VAR(1) with matrix Φ; λ_i: the loadings of series i, stacked in Λ
  - e_{it}: idiosyncratic AR(1) errors with coefficients ρ_i; u_t, ν_{it}: shocks
  - state $(f_t', f_{t-1}', e_t')'$; measurement matrix $Z = (\Lambda, 0, I)$
- Mixed frequencies Mariano and Murasawa (2003)
  - a quarterly growth rate is a weighted sum of monthly states
  - missing months are simply skipped by the filter

The DFM as a state space model (2/2)

- Estimation
  - two steps Doz, Giannone and Reichlin (2011): principal components give Λ, Φ, Σ_e; then the Kalman smoother re-estimates f_t on the full panel; consistent as $N, T \to \infty$
  - ML by EM Shumway and Stoffer (1982), Bańbura and Modugno (2014): the E-step is the smoother, arbitrary missing patterns are allowed
  - large N: collapse the N observations into an r-dimensional vector before filtering Jungbacker and Koopman (2015); the cost no longer grows with N^3
- Nowcasting Giannone, Reichlin and Small (2008): the ragged edge, news decomposition and real-time evaluation are in Chapter 5; this section shows the state space engine

A one-factor model of the US coincident indicators



- Monthly growth of industrial production, payrolls, real income less transfers and real sales (FRED), standardised, January 1990–September 2026; two-step estimator, univariate treatment of the observation vector; last 36 months

ATS_ch6_dfm_ragged_edge

Interpreting the ragged edge

- Loadings 0.90, 0.89, 0.68, 0.84; factor AR coefficient 0.12: monthly growth rates have little persistence, so the factor is mostly a weighted average of the month
- The last complete month is July 2026: the factor s.d. is 0.28; in September 2026, with one series of four available, it rises to 0.46
- The filter weighs each series by its signal-to-noise ratio; no series is dropped, no gap is filled by hand
- Forecasting the factor beyond the data is the same recursion with all observations missing

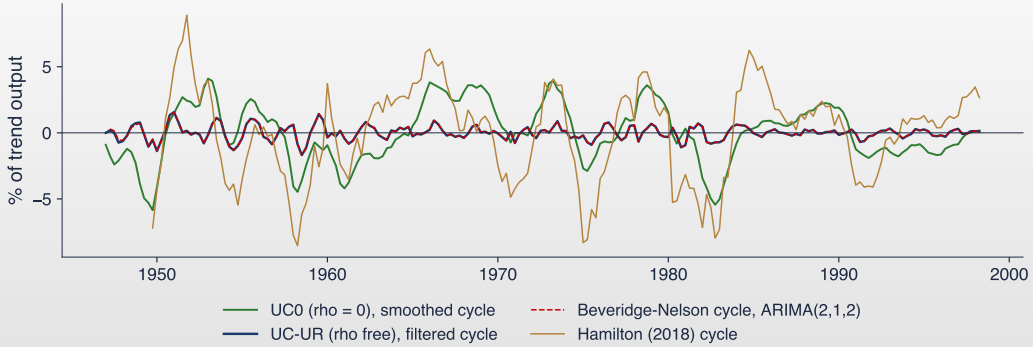
Three definitions of the cycle

- Beveridge–Nelson Beveridge and Nelson (1981): trend = long-horizon forecast net of drift
  - $\tau_t^{BN} = y_t + \sum_{j \geq 1} \mathbb{E}_t(\Delta y_{t+j} - \mu)$; cycle $c_t^{BN} = y_t - \tau_t^{BN}$
  - $\mathbb{E}_t$: expectation given data to t; μ: mean growth; the trend is where y is expected to settle once the predictable growth has died out
  - for an AR(1) in growth: $c_t^{BN} = -\frac{\phi}{1-\phi}(\Delta y_t - \mu)$; one-sided, depends on the ARMA chosen; reliable versions: Kamber, Morley and Wong (2018)
- Unobserved components Clark (1987), Watson (1986), Harvey and Jaeger (1993): $y_t = \tau_t + c_t$ with a random-walk (or smooth) trend and an AR(2) cycle; two-sided after smoothing
- Hamilton Hamilton (2018): c_{t+h} = residual of y_{t+h} on $(1, y_t, \ldots, y_{t-3})$, $h = 8$ quarters; model-free and one-sided (TSA, Chapter 10)
- For US GDP the BN cycle is small and noisy, the UC cycle large and smooth: the same data, two stories. Why?

Unobserved components with correlated shocks

- Morley, Nelson and Zivot (2003), the UC-UR model
  - $y_t = \tau_t + c_t$, $\tau_t = \mu + \tau_{t-1} + \eta_t$, $c_t = \phi_1 c_{t-1} + \phi_2 c_{t-2} + \varepsilon_t$, $\text{Corr}(\eta_t, \varepsilon_t) = \rho$
  - τ_t: the stochastic trend with drift μ and shocks η_t (s.d. σ_η); c_t: an AR(2) cycle with shocks ε_t (s.d. σ_ε); ρ: the correlation of the two shocks
- Clark's UC0 imposes $\rho = 0$; with an AR(2) cycle, ρ is identified: the reduced form is an ARIMA(2,1,2) with as many parameters as UC-UR
  - with an AR(1) cycle, ρ is not identified
- Result: if ρ is free, the filtered UC cycle equals the BN cycle of the ARIMA(2,1,2): the difference between BN and UC was the restriction $\rho = 0$
- State (τ_t, c_t, c_{t-1}): τ diffuse, the cycle block stationary (mixed initialisation); case study below: their sample, 1947Q1–1998Q2, today's data vintage

Replicating Morley, Nelson and Zivot (2003)



- 100×log US real GDP (FRED GDPC1), $T = 206$; UC0 and UC-UR by exact diffuse ML; BN cycle from an ARIMA(2,1,2); Hamilton (2018) cycle for comparison

ATS_ch6_trend_cycle

Interpreting the MNZ replication

Model	μ	ϕ_1	ϕ_2	σ_η	σ_ε	ρ	$\ln L$
UC0	0.858	1.501	−0.571	0.612	0.665	0	−280.81
UC-UR	0.859	1.334	−0.738	1.185	0.669	−0.927	−279.35

- $\hat{\rho} = -0.927$: trend and cycle shocks are almost perfectly negatively correlated; trend shocks are larger than cycle shocks ($\hat{\sigma}_\eta > \hat{\sigma}_\varepsilon$)
- Correlation of the UC-UR filtered cycle with the BN cycle (ARIMA AR roots 1.334, −0.738 as in UC-UR): 0.999; s.d. of the cycles: UC0 2.20, UC-UR 0.52, BN 0.52, Hamilton 3.50
- LR test of $\rho = 0$: 2.92, $p = 0.088$ (χ^2_1); the cycle of UC0 is large because it is assumed orthogonal to the trend, not because the data demand it
- Our numbers use today's GDP vintage; the paper's estimates differ in the decimals, not in the conclusion

Trend shocks and the cycle in Romanian GDP

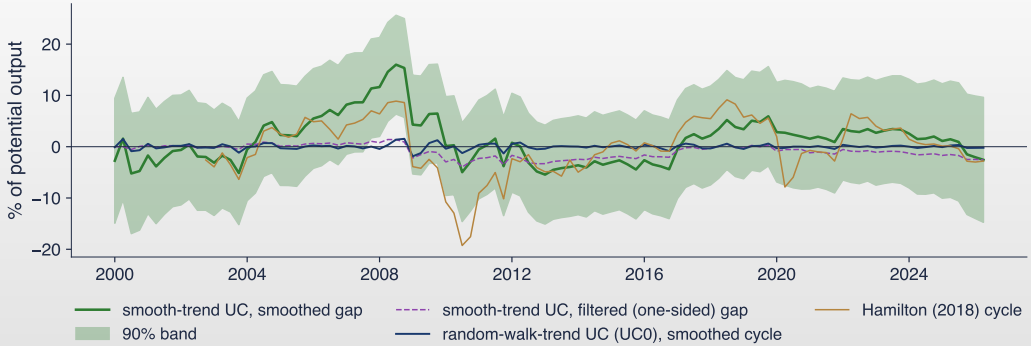


The BNR palace, Bucharest

Photo: Ștefan Jurcă (2015); CC BY 2.0; Wikimedia Commons

- Emerging economies have volatile trend shocks Aguiar and Gopinath (2007): a random-walk trend absorbs most of the movement in GDP
- Romanian UC0, 2000Q1–2026Q2 (2020Q2–Q4 treated as missing): $\hat{\sigma}_\eta = 1.897$, $\hat{\sigma}_\varepsilon = 0.950$, $\hat{\phi}_1 = 0.280$, $\hat{\phi}_2 = -0.261$: a short, weak cycle
- UC-UR: $\hat{\rho} = 1.00$ at the boundary, LR $= 2.92$: in 106 quarters the correlation is poorly determined
- Alternative: a smooth trend (integrated random walk, the HP model Harvey and Jaeger (1993)) with an AR(2) cycle: $\hat{\sigma}^2_\zeta = 0.0046$, $\hat{\sigma}^2_\varepsilon = 4.95$, $\hat{\phi} = (0.87, 0.09)$

The Romanian output gap



- 100×log real GDP, Eurostat, seasonally and calendar adjusted; smooth-trend UC (smoothed, with 90% band, and filtered), random-walk-trend UC0, Hamilton (2018) on the full series

ATS_ch6_trend_cycle

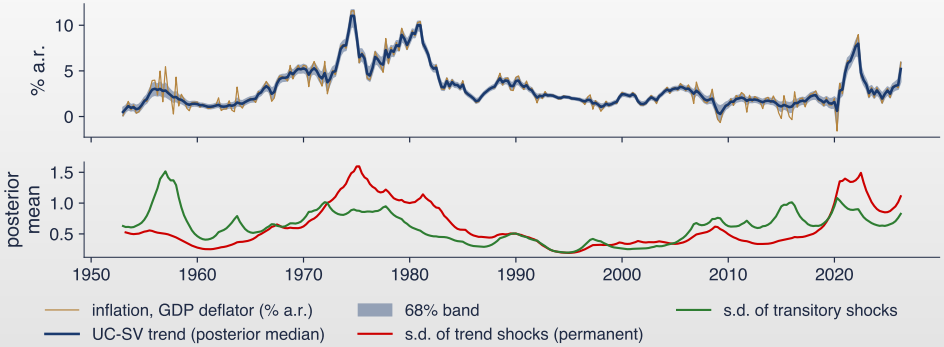
Interpreting the Romanian output gap

- Smooth-trend gap: 16.0% in 2008Q3 (s.d. 5.9), −5.0% in 2010Q3, 5.9% in 2019Q4; last quarter −2.6% (s.d. 7.4)
- Real time is different: the filtered gap was only 1.4% in 2008Q3 and 0.3% in 2019Q4; s.d. of the revisions (smoothed minus filtered): 3.8 pp Orphanides and van Norden (2002)
- Model choice dominates: s.d. of the gap 4.5 (smooth trend), 0.53 (UC0), 5.4 (Hamilton); correlation smooth trend–Hamilton 0.65
- A Romanian output gap without its band and without the model behind it is not information

The UC-SV model of Stock and Watson (2007)

- $\pi_t = \tau_t + \sigma_{\eta,t}\eta_t, \quad \tau_t = \tau_{t-1} + \sigma_{\varepsilon,t}\varepsilon_t, \quad \ln \sigma^2_{j,t} = \ln \sigma^2_{j,t-1} + \nu_{j,t},\ \nu_{j,t} \sim N(0,\gamma)$ Stock and Watson (2007)
  - π_t: inflation; τ_t: trend inflation (a random walk); $\sigma_{\eta,t}$, $\sigma_{\varepsilon,t}$: the time-varying s.d. of the transitory and trend shocks ($j \in \{\eta, \varepsilon\}$); $\eta_t, \varepsilon_t \sim N(0,1)$
- $\gamma = 0.2$ is fixed, not estimated: the only tuning parameter; it governs how fast the volatilities can move
  - reduced form: IMA(1,1) with a time-varying MA coefficient θ_t, a function of $\sigma_{\varepsilon,t}/\sigma_{\eta,t}$
- Gibbs: $\tau_{1:n}$ by the precision sampler (time-varying weights); each $\ln \sigma^2_{j,1:n}$ by the KSC mixture on $\ln(\hat{\eta}_t^2 + c)$ and $\ln(\Delta\tau_t^2 + c)$
  - for Romania: three quarterly seasonal coefficients as an extra Gibbs block (weighted regression); the HICP is not seasonally adjusted
- Why it matters: the share of trend shocks decides how much of an inflation surge is permanent and how much a forecaster should extrapolate

US trend inflation, 1953–2026



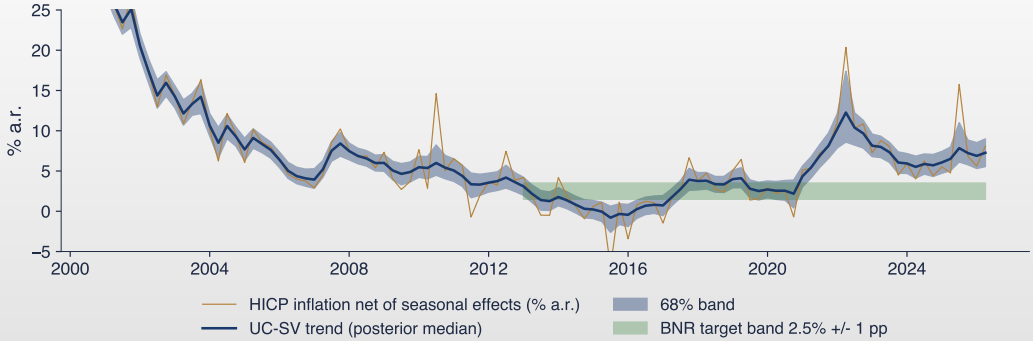
- GDP-deflator inflation, $T = 294$ quarters; UC-SV with $\gamma = 0.2$, Gibbs sampler; top: trend with 68% band; bottom: posterior means of $\sigma_{\varepsilon,t}$ (trend) and $\sigma_{\eta,t}$ (transitory)

ATS_ch6_ucsv_inflation

Interpreting US trend inflation

- Trend: 8.9% in 1975Q1, 2.1% in 1995Q1, 1.4% in 2019Q4, 8.0% [6.6; 8.9] in 2022Q2, 5.2% [3.8; 6.0] in 2026Q2
- Trend shocks: s.d. 1.60 in 1975, 0.19 in 1995, 1.42 in 2022; transitory shocks: 0.88, 0.20, 0.90
- Implied MA coefficient θ_t: -0.20 (1975), -0.39 (1995), -0.37 (2019), -0.24 (2022): inflation became more like a random walk again in 2021–2022
- This is the forecasting lesson of Stock and Watson (2007): in calm periods smooth heavily; when trend volatility rises, a fixed-coefficient model reacts too slowly

Romanian trend inflation and the BNR target



- Quarterly HICP inflation (quarterly averages of the monthly index), $T = 102$, 2001Q1–2026Q2; UC-SV with $\gamma = 0.2$ and quarterly seasonal dummies; inflation-target band of the BNR since 2013

ATS_ch6_ucsv_inflation

Interpreting Romanian trend inflation

- Trend: 8.4% when inflation targeting began (2005Q3), −0.8% in 2015Q3, 2.5% in 2019Q4, 8.1% in 2023Q1
- Last quarter: 7.3% [5.5; 9.0]; posterior probability that the trend lies inside 1.5–3.5%: 2%
- Since 2013 the trend has been inside the band only part of the time: below it in 2015–2016 (VAT cuts), above it since 2021
- Caveats: administered prices (energy caps, VAT changes) enter as trend shocks in a univariate model; a project can add them as regressors or as a separate component

Extensions: BSTS and regime switching

- Bayesian structural time series Scott and Varian (2014): local linear trend + seasonal + regression with a spike-and-slab prior on many regressors (Google Trends)
  - the same simulation smoother inside a Gibbs sampler; variable selection on the regression block
  - counterfactuals for policy evaluation (CausalImpact, Brodersen et al. (2015)): Chapter 14
- Discrete states: $s_t \in \{1, \dots, K\}$ with a Markov chain; the Hamilton filter is the Bayes filter with sums instead of integrals
  - state space models with regime switching (Kim filter, Gibbs sampling) Kim and Nelson (1999): Chapter 7
- Forecast evaluation of all these models (density forecasts, scoring rules): Chapter 1

An open question

- How much of the 2021–2023 inflation surge was a shift in trend inflation, in the US and in Romania, and has it been reversed?
  - formal: the posterior of τ_t and of $\sigma_{\varepsilon,t}/\sigma_{\eta,t}$ in UC-SV; falsified if the trend has returned to its 2019 level with high posterior probability under every pre-registered specification
  - the univariate answer depends on one fixed parameter, γ: the mini-case below
- Why it matters: a central bank reacts differently to a transitory and to a persistent surge; anchoring is a statement about the trend, not about headline inflation
  - literature to start from: Stock and Watson (2007), Cogley and Sargent (2005), Primiceri (2005), Morley, Nelson and Zivot (2003)

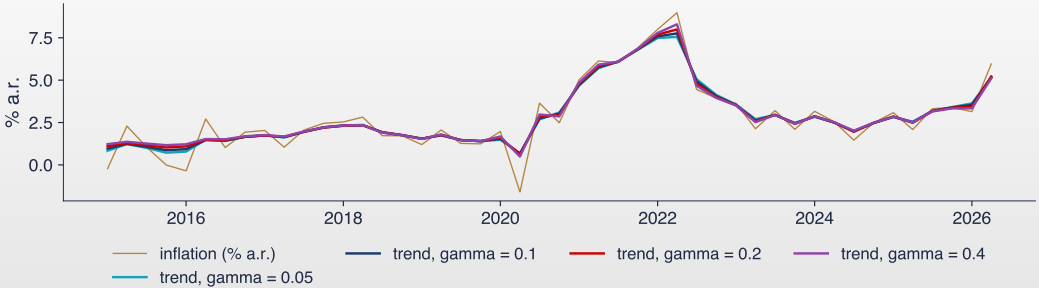
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed papers that estimate trend inflation with stochastic volatility for European countries after 2021; give DOIs.` Then check every DOI on Crossref
  - **hypothesis**: `Propose three observable implications that distinguish a trend shift from a sequence of large transitory shocks.`
  - **code and replication**: ask for a UC-SV Gibbs sampler, then reproduce a known number first (the US trend of this lecture)
  - **robustness and critique**: `Act as a hostile referee: which fixed choices of a UC-SV model could change the conclusion about 2022?`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

Required checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The sampler targets the stated posterior: test it on a model with a known answer (the Kalman smoother, as in the three-sampler check)
- Convergence is reported: several chains, inefficiency factors, the effect of doubling the draws
- Fixed choices (γ, the KSC offset, priors, seasonal treatment, start date) are listed and varied before conclusions are drawn
- End-of-sample estimates are reported as filtered values with their bands, not as smoothed point estimates

Mini-case: sensitivity of the UC-SV trend to a fixed parameter



- US UC-SV trend since 2015 for $\gamma \in \{0.05, 0.1, 0.2, 0.4\}$: trend in 2022Q2 7.6, 7.8, 8.0, 8.3% (inflation 9.0%); in the last quarter 5.1, 5.2, 5.2, 5.1%
- The ranking is stable but the level of the 2022 trend is not: an AI summary that reports "trend inflation peaked at X%" without γ hides a modelling choice

ATS_ch6_ai_robustness

Project idea

- **Trend inflation and anchoring in Central and Eastern Europe**: replicate first, then extend
  - replicate: the US UC-SV trend of this lecture (2022Q2: 8.0%) and the Romanian trend (7.3% in 2026Q2)
  - extend: Romania, Hungary, Poland and Czechia; a multivariate UC-SV with a common euro-area trend; administered prices as a separate component; γ estimated with a prior instead of fixed
  - pre-register: data, sample, priors, the anchoring statistic (posterior probability inside the target band), the real-time exercise and the robustness grid
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- The Kalman filter is Gaussian conditioning applied recursively; the smoother, the likelihood and the simulation smoother follow from it
- Initialise nonstationary states exactly; treat zero variances as boundary problems
- Bayesian state space: sample whole paths; FFBS, DK and precision samplers are interchangeable
- Stochastic volatility is not GARCH; the KSC mixture and particle filters make it tractable
- Latent quantities (gaps, trends, time-varying slopes) come with model-dependent uncertainty: report bands, filtered values and alternatives

Self-assessment

Questions

- Why does a big-kappa likelihood fall without bound as κ grows?
- What does an ML estimate $\hat{\sigma}_\eta^2 = 0$ tell you about a local level?
- Why does the DK simulation smoother need only a mean smoother?
- Why is the EKF gain zero in the SV model?
- Why is $\ln \hat{L}$ biased although $\hat{L}$ is unbiased?
- Which restriction makes the UC cycle differ from the BN cycle?

Next:

- Regime-switching models
- Discrete hidden states: the Hamilton filter, the Kim smoother, EM and Gibbs for Markov switching, and state space models with regimes

Appendix: the smoother from the lemma

- $Y_n = (Y_{t-1}, v_t, \ldots, v_n)$, and $v_t, \ldots, v_n$ are independent of each other and of Y_{t-1}: the lemma applies to each v_j separately
- Smoother: $\hat{\alpha}_t = a_t + \sum_{j=t}^{n} \mathrm{Cov}(\alpha_t, v_j) F_j^{-1} v_j$, with $\mathrm{Cov}(\alpha_t, v_j) = P_t L_t' \cdots L_{j-1}'$
- Hence $\hat{\alpha}_t = a_t + P_t r_{t-1}$ with $r_{t-1} = \sum_{j=t}^{n} L_t' \cdots L_{j-1}' Z_j' F_j^{-1} v_j$ (empty product for $j = t$), which satisfies the backward recursion $r_{t-1} = Z_t' F_t^{-1} v_t + L_t' r_t$
- The filtering steps (update and prediction) follow from the same lemma with $j = t$ only: slide Derivation of the Kalman filter from a Gaussian lemma

Back

Appendix: exactness of the Durbin–Koopman simulation smoother

- Gaussian model: $\alpha | y \sim N(\hat{\alpha}(y), V)$ with V independent of y, and $\hat{\alpha}(y)$ linear in y (affine with the intercepts)
- For a draw (α^+, y^+) from the joint law, $\alpha^+ - \hat{\alpha}(y^+) \sim N(0, V)$ and is independent of y^+ (and of y)
- Hence $\tilde{\alpha} = \hat{\alpha}(y) + \alpha^+ - \hat{\alpha}(y^+) \sim N(\hat{\alpha}(y), V)$; linearity gives $\hat{\alpha}(y) - \hat{\alpha}(y^+) = \hat{\alpha}_0(y - y^+)$, one smoother pass with zero intercepts
- Diffuse states: the exact diffuse smoother is invariant to the diffuse part of α_1^+, so it can be set to a_1

Back

Appendix: unbiasedness of the particle likelihood

- Bootstrap filter, one step: $\mathbb{E}[\frac{1}{N}\sum_i w_t^{(i)} \mid \mathcal{F}_{t-1}] = \int g(y_t|x)\,\hat{p}_N(x|Y_{t-1})\,dx$, with $\hat{p}_N$ the empirical predictive law of the particles
- Applying the tower property successively for $t = n, n-1, \ldots, 1$ (at each step the expectation of the last factor given $\mathcal{F}_{t-1}$ is replaced by the formula above), with multinomial or systematic resampling, gives $\mathbb{E}\prod_t \hat{p}(y_t|Y_{t-1}) = p(y_{1:n})$ Del Moral (2004)
- Each factor is a ratio estimate, but the product is unbiased because resampling preserves the expected weights; normalised quantities (filtered means) are only consistent

Back

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